TASFIA NOOR CHOWDHURY



Mechatronics Engineer | Robotics & AI Researcher | Educator

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ResearchGate · Website · Portfolio

PROFILE

Mechatronics engineer and robotics researcher pursuing an M.Sc. in Robotics & Mechatronics Engineering at the University of Dhaka (B.Sc., CUET; senior-year GPA 3.96/4.00; five-time Dean's Award). I build systems that let machines learn from people: robots that learn a manipulation skill from a single hand-gesture demonstration, mobile robots that learn safe warehouse navigation from human driving, and deep-learning models for medical imaging and flood mapping. Six papers, including two presented at IEEE RAAICON 2026 and an ICML 2026 workshop poster. Hands-on with ROS 2, MuJoCo, PyTorch, and PLC automation (Siemens TIA Portal, Factory I/O), with five years of STEM teaching and leadership experience as an Aspire Leaders Fellow and e-commerce founder. Looking for a robotics, AI/ML, or automation engineering role in Japan.

RESEARCH INTERESTS

Robot Learning & Intelligent Systems · Machine Learning & Deep Learning · Computer Vision & Medical Image Analysis · Data Science & Big Data Analytics · Industrial Automation · IoT & Embedded Systems

EDUCATION

M.Sc. in Robotics & Mechatronics Engineering, University of Dhaka — Jul 2026 – Jun 2027 (expected) · Dhaka

- Thesis: *Knee Abnormality Detection from MRI Images* (RSNA dataset), supervised by Avishek Kumar Ghosh, MAIM Lab
- Courses: Neural Networks & Deep Learning, Industrial Automation, Automotive Control & Simulation, Engineering Research Methodology

B.Sc. in Mechatronics & Industrial Engineering, Chittagong University of Engineering and Technology (CUET) — Feb 2019 – Jun 2024 · Chattogram

- CGPA 3.63/4.00 (76.60%) · Class rank 8th of 30 · Year-wise GPA: Sophomore 3.75 · Junior 3.76 · Senior **3.96**
- Dean's Award 5 consecutive times (GPA 3.75+) · Dutch-Bangla Bank Foundation Scholarship, monthly stipend (2019–2024)
- Thesis: *A Hybrid Data-Driven Approach for Analyzing and Predicting Inpatient Length of Stay in Health Care Centers*, supervised by Kazi Naimur Rahman

Higher Secondary Certificate (HSC), Science, Barisal Govt. Women's College · GPA 5.00/5.00 · Government Scholarship — Apr 2016 – Jun 2018

Secondary School Certificate (SSC), Barisal Govt. Girls High School · GPA 5.00/5.00 · Government Scholarship — Jun 2016

RESEARCH EXPERIENCE

M.Sc. Thesis Researcher, MAIM Lab, University of Dhaka — Jul 2026 – Present · Dhaka

- Developing deep-learning models to detect knee abnormalities from MRI scans using the RSNA dataset

Independent Researcher, Medical Robotics Lab, FAU Erlangen-Nürnberg (Germany), under M. A. Hafiz — Jul 2025 – Jun 2026 · Remote

- Conducted solo research on robot learning in MuJoCo physics simulation
- Built learning-from-demonstration pipelines for robot manipulation and mobile-robot navigation, leading to two IEEE RAAICON 2026 papers

B.Sc. Thesis Researcher, Dept. of Mechatronics & Industrial Engineering, CUET — 2023 – 2024 · Chattogram

- Built a big-data pipeline to predict inpatient length of stay from 2.3 million hospital records

PUBLICATIONS

Conference Papers

1. Hand-Gesture-Guided One-Demo-to-Many Robot Skill Generation with Teacher-Weighted Residual Correction in MuJoCo

2026 IEEE 5th International Conference on Robotics, Automation, Artificial Intelligence and Internet of Things (RAAICON), Jashore University of Science and Technology · 3rd author

- Built a webcam hand-gesture interface (hand-landmark tracking → Cartesian end-effector commands) to teleoperate a simulated dual Franka Panda robot in MuJoCo
- Turned **one** recorded demonstration into **50** spatially varied, physics-validated demonstrations; 100% replayed successfully
- Geometry-preserving augmentation lifted behavior-cloning success to **98% / 96%** versus **48% / 26%** with naive noise (5 seeds, paired Wilcoxon p = 0.031)

2. Human-Demonstration-Guided Warehouse Navigation with Safety-Aware Gap Reasoning for Mobile Robots

2026 IEEE RAAICON, Jashore University of Science and Technology · 4th author

- Learned local navigation for a TurtleBot3 Burger in a Webots warehouse (ROS 2 Jazzy) from human teleoperation demos with 2D LiDAR, odometry, and dynamic-obstacle risk features
- MLP behavior-cloning policy (17-D state → velocity commands) with a rule-based safety layer: gap reasoning, GO / SLOW GO / WAIT decisions, wall avoidance, corner recovery
- Safety-aware rule layer cut close-obstacle exposure from **65% to 2.4%** of time; the learned policy cut recoveries by **40.8%** (90 runs of 600 s); built a live monitoring dashboard

Workshop Paper

Generative Traffic Forecasting: Preserving Shockwave Topology with Diffusion Models

M. I. Hoque, M. I. Khalil, **T. N. Chowdhury**, T. B. H. Jerin · *ICML 2026 Workshop on Forecasting (Forecast@ICML26), poster, 2026* · openreview.net/forum?id=KA5RdNascV

- DDPM with a history-aware conditional UNet1D on PeMSD7 and ensemble-medoid inference: MAE 5.32 mph, RMSE 9.76 mph, preserving shockwaves and phantom jams that MSE-trained baselines smooth out

Journal Article

3. Machine Learning Models for Optimizing Online Order Fulfillment: Forecasting Lead Time and Late Delivery Risk

N. Sultana, A. M. Moosa, S. M. F. Faisal, **T. N. Chowdhury** · *Journal of Optimization and Supply Chain Management, 2(2), 70–81, 2025* · doi.org/10.22034/ISS.2025.9373.1049

- Compared Linear Regression, Decision Tree, Random Forest, and XGBoost on 180,000+ e-commerce orders (DataCo Smart Supply Chain)
- XGBoost: lead time **R^2 0.70, RMSE 0.88 days**; late-delivery risk **accuracy 0.89, precision 0.92, recall 0.89, F1 0.90**; shipping method identified as the top predictor

Preprints

4. AI-Driven Water Segmentation with Deep Learning Models for Enhanced Flood Monitoring

S. A. Mou, **T. N. Chowdhury**, A. I. Mannan, S. N. Mim, L. Tarannum, T. Noman, J. U. Ahamed · *arXiv, 2025* · doi.org/10.48550/arXiv.2501.08266

- Role: conceptualization, methodology, formal analysis, validation
- Built a flood dataset by augmenting benchmarks with drone, in-field, and social-media images; compared segmentation models in PyTorch: **DeepLab v3 90.57%**, ResNet 88.70%, U-Net 87.12% validation accuracy

5. A Hybrid Data-Driven Approach for Analyzing and Predicting Inpatient Length of Stay in Health Care Centers (B.Sc. thesis)

T. N. Chowdhury, S. A. Mou, K. N. Rahman · *arXiv, 2025* · arxiv.org/abs/2501.18535

- Analyzed **2.3 million** de-identified New York inpatient records (SPARCS) with Apache Spark on AWS; PCA to find the top factors driving length of stay
- Predicted length of stay at admission with Decision Tree, Logistic Regression, Random Forest, AdaBoost, and LightGBM (cross-validated); combined ML with process mining and simulation to propose patient-flow improvements

Technical report: Report on Industrial Training at Berger Paints Ltd, Chittagong Factory · ResearchGate

PROFESSIONAL EXPERIENCE

Communication & Technical Lead, ILM International School — Jul 2023 – Present · Remote

- Led communication strategy and design workflows, contributing to a **25%** increase in student enrollment
- Oversaw the technical side of the school's digital presence and coordinated design and marketing teams for a consistent brand
- Analyzed engagement metrics to refine outreach campaigns; streamlined communication between administration and faculty

STEM Instructor & Academic Mentor, Freelance — Jan 2021 – Dec 2025 · Remote & On-site

- Delivered customized Physics, Mathematics, and IT lessons, adapting methods to different learning speeds
- Mentored students to significant academic improvement by simplifying complex engineering and scientific concepts

Business Evangelist, ILM Planet — Jul 2022 – Jan 2026 · Remote

- Executed strategic and push-marketing plans, raising lead generation and customer retention by **20%**
- Administered payroll, contracts, and customer relations using MS Excel

Founder, Noor Group (e-commerce) — 2021 – Present

- Founded and run an online shop for books, clothing, and daily essentials; Community Startup Award for exceptional growth (2021)

INDUSTRIAL TRAINING

Industrial Induction, Berger Paints Bangladesh Ltd, Chittagong Factory (Kalurghat) — Aug – Sep 2023 · 3 weeks

- Studied vertical paint-manufacturing processes and machinery; assisted Quality Control in monitoring compliance with industry standards

Roadmap to ISO 45001:2018, EHS training by Marico Bangladesh

- Occupational health & safety management; led Hazard Identification & Risk Analysis (HIRA) and operational-control simulations

Industrial Visit, PepsiCo BD, Chittagong Factory

- Analyzed manufacturing-line workflows and applied the findings to process-optimization coursework

PROJECTS

EV Battery-Charging Stations for Electric Three-Wheelers: Winner, Solve-a-thon 2025 (EU, World Bank, WePower, Devtale Partners) — 2025

- Designed, as part of a team, charging infrastructure for the electric three-wheelers widely used in Bangladesh

Automated Box-Sorting Station (Factory I/O, PLC Ladder Diagram)

- Programmed a PLC sorting line with a vision sensor and pusher; tuned conveyor speed and sensor–actuator synchronization for error-free sorting

Eye-Gaze Controlled Wheelchair: 2nd Runner-Up, Project Idea Competition — 2022

- Designed an assistive wheelchair that people with mobility impairments can steer using eye gaze

IoT Smart Irrigation System (Arduino, sensors): Top 250, Unibator BD — 2021

- Automated sprinkler system using real-time soil and weather data to conserve water

TECHNICAL SKILLS

Programming	Python (NumPy, Pandas), C, MATLAB, Simulink
Machine Learning & AI	PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM · Deep Learning (U-Net, ResNet, DeepLab v3) · Computer Vision · Imitation Learning / Behavior Cloning · Classical ML (Regression, Decision Tree, Random Forest, AdaBoost) · PCA · Medical Image Analysis (MRI) · LLMs
Robotics & Simulation	ROS 2 (Jazzy, Nav2), MuJoCo, Webots, Gazebo · UR3 cobot, TurtleBot3, Franka Panda (sim), De Lorenzo CIM-C 1200 · LiDAR & odometry · Teleoperation · 3D Printing
Industrial Automation	Siemens TIA Portal, Factory I/O, PLC Ladder Diagram, Function Block Diagram
Electronics	Arduino Uno, Microcontrollers, Sensors, Proteus, Tinkercad
Data & Cloud	Apache Spark, AWS, Power BI, Advanced Excel, Matplotlib, Seaborn, Plotly
CAD & Design	AutoCAD, SolidWorks, Figma
Industrial & Quality	Six Sigma, Lean, Quality Control, ISO 45001:2018, HIRA
Tools	Jira, MS Office, Mendeley, ResearchRabbit, PRISMA

Soft Skills	Leadership & team management · Communication & presentation · Mentoring · Cross-functional coordination · Analytical problem solving · Independent research · Remote collaboration

LEADERSHIP PROGRAMS & CERTIFICATIONS

- **Fellow**, Aspire Leaders Program 2025 (Harvard Business School initiative)
- **Graduate**, Shoktikonna Leadership Cohort 2025 (EU, GIZ, World Bank, WePower, Devtale Partners): four-month leadership training on sustainable energy, including the Solve-a-thon and Green Skills Summit
- Leading Teams: Building Effective Team Cultures · Leading Teams: Developing as a Leader · Introduction to Business Analytics: Communicating with Data · Sustainability Reporting: From Regulation to Implementation, University of Illinois Urbana-Champaign (Coursera), Nov 2025
- Technical Foundations of Energy: Transmission Systems, World Bank Open Learning Campus, Nov 2025

AWARDS & HONORS

- Winner, Solve-a-thon 2025 (EU, World Bank, WePower, Devtale Partners)
- Dean's Award, 5 consecutive times, CUET · Dutch-Bangla Bank Foundation Scholarship (2019–2024)
- Best Dining Manager Award for outstanding management, Shamsen Nahar Khan Hall, CUET (2024)
- 2nd Runner-Up, Project Idea Competition, Eye-Gaze Controlled Wheelchair (2022)
- Community Startup Award for exceptional growth, Noor Group (2021) · Top 250, Unibator BD (2021)
- Semi-Finalist, Hult Prize @ CUET, Team Bottle Green House (2019)
- Regional Winner, English Olympiad & Physics Olympiad (2017) · Government Scholarships for JSC, SSC, and HSC results

RELEVANT COURSEWORK

Robotics & Control	Robotics, Robotics System Design, System Dynamics & Control, Control Theory, Digital Control Systems, Industrial Automation, Automotive Control & Simulation
Electrical & Electronics	Electrical Circuits, Electrical Machines, Power Electronics & Drives, Digital Logic Design, Sensors & Instrumentation
Data Science	Machine Learning, Neural Networks & Deep Learning, Signals & Systems
Manufacturing & Management	CAD/CAM, Computer Integrated Manufacturing, Mechatronics & Industrial System Design, Reliability & Quality Control, Operations Management, Supply Chain, Materials Handling & Maintenance

LANGUAGES & TEST SCORES

Bengali (Native) · English (Professional) · IELTS Academic 6.5 (Listening 8.0, Reading 6.5, Writing 6.0, Speaking 6.0), Dec 2024

AFFILIATIONS

General Member, Robo Mechatronics Association (RMA), CUET · General Member, Industrial Engineering and Operations Management (IEOM) Society, CUET Chapter

REFERENCES

Available on request (CUET faculty, Dept. of Mechatronics & Industrial Engineering, including B.Sc. thesis supervisor).